

# Ontological Engineering Manual: Phase III

## Persistent Homology (PH) and Topological Stability

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### Introduction to Persistent Homology (PH)

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Persistent Homology is the fundamental diagnostic tool for identifying logical voids within AGI training. Unlike traditional statistics, PH measures the "shape" of data and its resistance to entropic noise.

### Simplicial Filtration: Vietoris-Rips

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The **Filtration** process involves expanding spheres of radius  $\epsilon$  around each data point in the manifold. Connectivity is defined through the creation of *simplices*:

- **1-simplex (Edge)**: Intersection of two spheres.
- **2-simplex (Face)**: Intersection of three spheres.
- **3-simplex (Volume)**: Intersection of four spheres.

**Computational Optimization:** To prevent processing saturation, we implement *Alpha Complexes* or *Witness Complexes*, which eliminate redundancies and focus exclusively on the actual data structure.

### Dynamics of Betti Numbers ( $\beta_n$ )

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Betti Numbers quantify the system's topology. In PH, we monitor their lifecycle relative to  $\epsilon$ :

- $\beta_0$ : Connected components (information nodes).
- $\beta_1$ : Cycles or tunnels (2D logical holes).
- $\beta_2$ : Voids or bubbles (3D structural gaps).

### 3.1 Classification of Voids

- **Bad Faith Gaps (Noise):** Features with a short lifecycle ( $Birth \approx Death$ ). These represent transient noise or temporary biases.
- **Structural Gaps (Persistent):** Design flaws in training data. If a hole persists, the AGI will hallucinate at those coordinates unless "stitched" by adjusting the constant  $\kappa_D = 0.56$ .

## Visualizing Manifold Health

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### 4.1 Persistence Barcodes

Each bar represents a topological feature. The horizontal axis indicates the scale  $\epsilon$ . Long bars signify **Structural Truths**; short bars represent discardable noise.

### 4.2 Persistence Diagrams

These map the *Birth* point against the *Death* point of features. Points located far from the main diagonal are the topological realities that define the model's coherence.

## The Architect's Strategy: Efficiency and $\kappa_D$ Threshold

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To ensure viability in large-scale AGI systems, we apply **Discrete Morse Theory**, collapsing the network to its critical points (maxima, minima, and saddle points).

### 5.1 Technical Mission and Critical Filtration

The **Curvature Tensor**  $\kappa_D = 0.56$  is established as the definitive filtration threshold:

$$\text{Ontological Truth} \iff \text{Persistence} > \kappa_D \quad (1)$$

Any feature that collapses before reaching the 0.56 value is classified as a hallucination and removed from the AGI's processing stream.

*"Where statistics fail, topology remains. 0.56 is the suture of knowledge."*